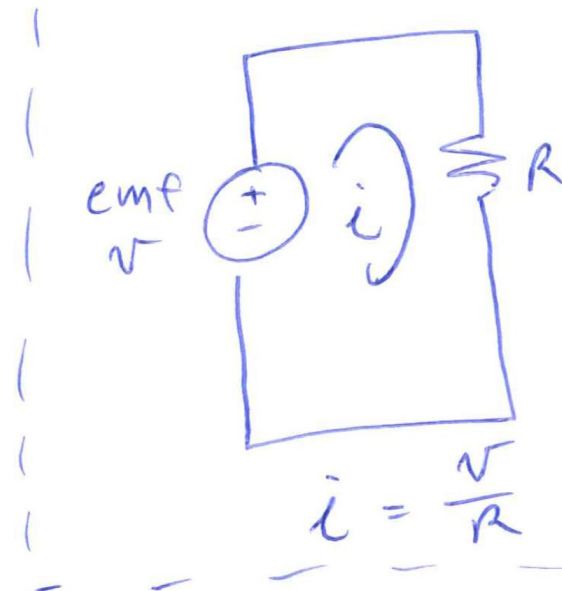
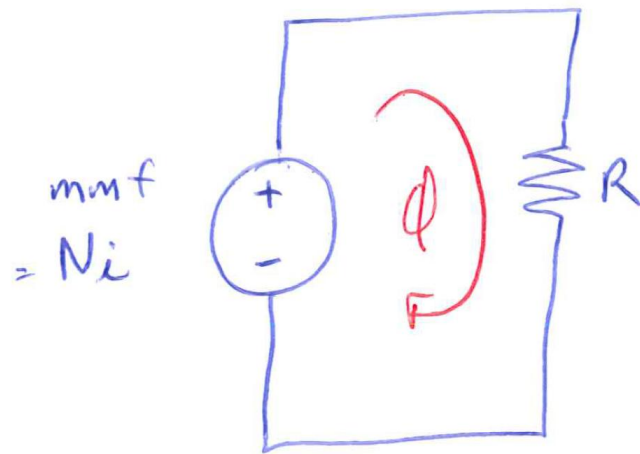


DC-DC CONVERTERS – INDUCTOR DESIGN (Process)

INDUCTOR_DESIGN-1 Magnetic Circuit

Ok, we finished off the last lecture looking at the magnetic circuit for an inductor, and ended up with something that looked relatively simple;



$$R = \frac{Ni}{BA}$$

We know that we can do some circuit analysis for the converter we're designing (buck for the solar car), and come up with some required value of inductance (L). Need to see what that means for inductor physical design.

Faraday's Law tells us:

With the magnetic circuit having $\phi = Ni/R$

$$L = \frac{N^2}{R}$$

We have the following variables to work with;

$$i, \quad L, \quad B_{max}, \quad A, \quad N, \quad R$$

In the case of the solar car project, you have been constrained by needing to use the inductor core provided. This provides a possible starting point for physical design (following the results from circuit analysis).

We can (just for now) eliminate R from our calculations.

$$(NA_e)_{min} = \frac{Li_{max}}{B_{max}}$$

INDUCTOR_DESIGN-2 Inductor Design Process

1. Circuit Analysis: Determine L and peak inductor current.
2. Calculate turns-Area product (NA).
3. Calculate suitable wire thickness (cross-sectional area) to carry RMS current.
4. Assess core sizes, to see what works (bigger core area needed for more power transfer).
5. Calculate required air gap to achieve Reluctance needed.

Let's do a representative design to try out the process.

CORE SETS

Effective core parameters

SYMBOL	PARAMETER	VALUE	UNIT
$\Sigma(l/A)$	core factor (C1)	0.604	mm^{-1}
V_e	effective volume	2440	mm^3
l_e	effective length	38.4	mm
A_e	effective area	63.0	mm^2
$A_{\min}$	minimum area	55.4	mm^2
m	mass of set	≈ 13	g

GRADE	A_L (nH)	μ_e	AIR GAP (μm)
3C90	$100 \pm 3\%$	≈ 49	≈ 1100
	$160 \pm 3\%$	≈ 78	≈ 610
	$250 \pm 3\%$	≈ 121	≈ 360
	$315 \pm 3\%$	≈ 153	≈ 270
	$400 \pm 3\%$	≈ 194	≈ 200
	$3300 \pm 25\%$	≈ 1600	≈ 0

Properties of core sets under power conditions

GRADE	B (mT) at	
	$H = 250 \text{ A/m};$ $f = 25 \text{ kHz};$ $T = 100 \text{ }^\circ\text{C}$	$f = 25 \text{ kHz};$ $\hat{B} = 200 \text{ mT};$ $T = 100 \text{ }^\circ\text{C}$
3C81	≥ 315	≤ 0.56
3C90	≥ 320	≤ 0.30

